

Lab 1:

```
14 (
15 select f.title, c.name as category, f.length,
16        dense_rank() over(
17          partition by c.name
18          order by f.length desc
19        ) as rank_
20 from film f
21 join film_category fc
22 on f.film_id = fc.film_id
23 join category c
24 on fc.category_id = c.category_id
25 )
26
27 select *
28 from ranked_films
29 where rank_ <= 3;
30
```

Data Output Messages Notifications

	title character varying (255)	category character varying (25)	length smallint	rank_ bigint
1	Worst Banger	Action	185	1
2	Darn Forrester	Action	185	1
3	Casualties Encino	Action	179	2
4	Quest Mussolini	Action	177	3
5	Pond Seattle	Animation	185	1
6	Gangs Pride	Animation	185	1
7	Theory Mermaid	Animation	184	2
8	Sons Interview	Animation	184	2
9	Lawless Vision	Animation	181	3
10	Wrong Behavior	Children	178	1
11	Fury Murder	Children	178	1
12	Empire Malkovich	Children	177	2
13	Model Fish	Children	175	3
14	Conspiracy Spirit	Classics	184	1
15	Dracula Crystal	Classics	176	2

Lab 2:

```

34 select date_trunc('month', payment_date) as month,
35 sum(amount) as revenue
36 from payment
37 group by month
38 )
39
40 select month, revenue,
41 lag(revenue) over(
42 order by month
43 ) as previous_revenue,
44
45 (revenue-lag(revenue) over(order by month))
46 /
47 lag(revenue) over(order by month)
48 *100 as growth
49
50 from monthly_sales;

```

Data Output Messages Notifications

	month timestamp without time zone	revenue numeric	previous_revenue numeric	growth numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

Lab 3:

```

51 --lab 3:
52 select customer_id, payment_date, amount,
53 sum(amount) over( partition by customer_id order by payment_date) as cumulative_spend
54
55 from payment
56 order by customer_id, payment_date;

```

Data Output Messages Notifications

Showing

	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_spend numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91
10	1	2007-03-02 16:30:04.996577	0.99	41.90
11	1	2007-03-17 11:06:20.996577	4.99	46.89
12	1	2007-03-18 02:25:55.996577	0.99	47.88
13	1	2007-03-19 08:23:42.996577	0.99	48.87
14	1	2007-03-19 12:25:20.996577	2.99	51.86
15	1	2007-03-21 22:02:23.996577	0.99	52.85

Lab 4:

```

62 join city ci on a.city_id = ci.city_id
63 join country co on ci.country_id = co.country_id
64 join payment p on c.customer_id = p.customer_id
65 group by c.customer_id, c.first_name, c.last_name, co.country
66 ),
67 ranked_customer as
68 (
69     select *,
70     rank() over(
71         partition by country
72         order by total_spent desc
73     ) as spending_rank
74     from customer_spending
75 )
76
77 select customer_id, first_name, last_name, country, total_spent
78 from ranked_customer
79 where spending_rank = 1;

```

Data Output Messages Notifications

Showing rows: 1 to 108

	customer_id integer	first_name character varying (45)	last_name character varying (45)	country character varying (50)	total_spent numeric
1	218	Vera	Mccoy	Afghanistan	67.82
2	176	June	Carroll	Algeria	151.68
3	320	Anthony	Schwab	American Samoa	47.85
4	528	Claude	Herzog	Angola	93.80
5	381	Bobby	Boudreau	Anguilla	99.68
6	560	Jordan	Archuleta	Argentina	129.71
7	41	Stephanie	Mitchell	Armenia	118.75
8	266	Nora	Herrera	Austria	113.74
9	334	Raymond	Mcwhorter	Azerbaijan	108.75
10	590	Seth	Hannon	Bahrain	108.76
11	21	Michelle	Clark	Bangladesh	146.68
12	144	Clara	Shaw	Belarus	189.60
13	431	Joel	Francisco	Bolivia	92.78
14	178	Marion	Snyder	Brazil	194.61
15	91	Lois	Butler	Brunei	107.66